VERSAL BASE CHANGE GIVES SMOOTHNESS OF THE 2-FIBRE PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2012-2036 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 2038-2085, the complete author proof (46 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_artin.json (pb=1896, stmt 2012-2036, proof 2038-2085).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 045G: sources/stacks-artin/pinned-original-source.tex, source lines 2085–2085. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XF: sources/stacks-artin/pinned-original-source.tex, source lines 1887–1896. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06HM: sources/stacks-artin/pinned-original-source.tex, source lines 2085–2085. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CXI: sources/stacks-artin/pinned-original-source.tex, source lines 1757–1763. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07WL: sources/stacks-artin/pinned-original-source.tex, source lines 570–590. Cited by the retained statement or proof but not part of the

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collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07WK: sources/stacks-artin/pinned-original-source.tex, source lines 539–548. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XG: sources/stacks-artin/pinned-original-source.tex, source lines 1976–1988. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02HX: sources/stacks-artin/pinned-original-source.tex, source lines 2085–2085. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06EG: sources/stacks-artin/pinned-original-source.tex, source lines 2085–2085. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S, $\mathcal{X}$, U, x, u_0 be as in Definition 07XF. Assume*

(1) *$\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces,*
(2) *Δ is locally of finite type (for example if $\mathcal{X}$ is limit preserving), and*
(3) *$\mathcal{X}$ has (RS).*

Let V be a scheme locally of finite type over S and let y be an object of $\mathcal{X}$ over V. Form the 2-fibre product

$$\begin{array}{ccc} \mathcal{Z} & \longrightarrow & (Sch/U)_{fppf} \\ \downarrow & & \downarrow{\scriptstyle x} \\ (Sch/V)_{fppf} & \xrightarrow{y} & \mathcal{X} \end{array}$$

Let Z be the algebraic space representing $\mathcal{Z}$ and let $z_0 \in |Z|$ be a finite type point lying over u_0. If x is versal at u_0, then the morphism $Z \to V$ is smooth at z_0.

Proof. (The parenthetical remark in the statement holds by Lemma 0CXI.) Observe that Z exists by assumption (1) and Algebraic Stacks, Lemma 045G. By assumption (2) we see that $Z \to V \times_S U$ is locally of finite type. Choose a scheme W, a closed point $w_0 \in W$, and an étale morphism $W \to Z$ mapping w_0 to z_0, see Morphisms of Spaces, Definition 06EG. Then W is locally of finite type over S and w_0 is a finite type point of W. Let $l = \kappa(z_0)$. Denote $z_{l,0}$, $v_{l,0}$, $u_{l,0}$, and $x_{l,0}$ the objects of $\mathcal{Z}$, $(Sch/V)_{fppf}$, $(Sch/U)_{fppf}$, and $\mathcal{X}$ over $\mathrm{Spec}(l)$ obtained by pullback to $\mathrm{Spec}(l) = w_0$. Consider

$$\begin{array}{ccccc} \mathcal{F}_{(Sch/W)_{fppf},l,w_0} & \longrightarrow & \mathcal{F}_{\mathcal{Z},l,z_{l,0}} & \longrightarrow & \mathcal{F}_{(Sch/U)_{fppf},l,u_{l,0}} \\ & & \downarrow & & \downarrow \\ & & \mathcal{F}_{(Sch/V)_{fppf},l,v_{l,0}} & \longrightarrow & \mathcal{F}_{\mathcal{X},l,x_{l,0}} \end{array}$$

By Lemma 07WL the square is a fibre product of predeformation categories. By Lemma 07XG we see that the right vertical arrow is smooth. By Formal Deformation Theory, Lemma 06HM the left vertical arrow is smooth. By Lemma 07WK

we see that the left horizontal arrow is smooth. We conclude that the map

$$\mathcal{F}_{(Sch/W)_{fppf},l,w_0} \to \mathcal{F}_{(Sch/V)_{fppf},l,v_{l,0}}$$

is smooth by Formal Deformation Theory, Lemma 06HM. Thus we conclude that $W \to V$ is smooth at w_0 by More on Morphisms, Lemma 02HX. This exactly means that $Z \to V$ is smooth at z_0 and the proof is complete. □